



Ali R. Vahdati

EVOLUTIONARY BIOLOGIST · COMPUTATIONAL BIOLOGIST

Zurich, Switzerland

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Key Skills

Analytical	Biological modeling, Machine learning, Statistics, Mathematical optimization, Agent-based modeling, Working with large genomic datasets, Population genetics
Programming	Julia, Python, R, Nim, LaTeX, MySQL, SQLite, Git, Continuous integration, Unit testing, Docker, Singularity
Visualization	Vega-lite, Matplotlib, Processing, Makie.jl, Principles of effective visualization
Soft	Resilience, Leadership, Commitment, Adaptability
Transferable	Teaching, Presentation, Writing, Time management, Project management
Languages	English (fluent), Persian (native), German (beginner)

Work Experience

Consultant

IBS center for climate physics, University of Pusan

- I built a model to analyze the effect of climate on human genetic diversity

Busan, South Korea

May and June 2022

Senior lecturer

Christoph Zollikofer group, Department of Anthropology, University of Zurich

- I study the ancient human dispersals and extinctions using simulations.

Zurich, Switzerland

Aug. 2020 - Current

Postdoctoral researcher

Christoph Zollikofer group, Department of Anthropology, University of Zurich

- I developed research projects and analyzed results using statistical tools and machine learning.

Zurich, Switzerland

Sep. 2017 - Aug. 2020

International collaborations

Modeler

University of Madeira

- I built a model for understanding metacommunity dynamics of complex life-cycles in exploited seascapes.

Madeira, Portugal

2022

Modeler

Centro de investigación Científica de Yucatán

- I built statistical models for understanding interaction strengths in food webs.

Mérida, Mexico

Oct. 2021

Education

University of Zurich

PhD in Evolutionary Biology

- Thesis: Genotype networks: convergent evolution, population size and adaptation.
- Published four scholarly papers.

Zurich, Switzerland

Jan. 2012 - Sep. 2017

University of Manchester

M.S. in Evolutionary Biology and Genomics

- Thesis 1: Identifying QTL and genomic imprinting effects on mice tissue phenotypes.
- Thesis 2: Analysis of replication timing on genetic variation in the human genome.
- Published one scholarly paper.

Manchester, UK

Sep. 2010 - Sep. 2011

Selected Peer-reviewed Papers

- EvoDynamics.jl: a framework for modeling eco-evolutionary dynamics** *Journal of Open Source Software*
R Vahdati A, Melián C J 2022
- Exploring Late Pleistocene hominin dispersals, coexistence and extinction with agent-based multi-factor models** *Quaternary Science Reviews*
R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE 2022
- Agents.jl: A performant and feature-full agent based modelling software of minimal code complexity** *Simulation*
Datseris G, R Vahdati A, DuBois T C 2022
- Drivers of Late Pleistocene human survival and dispersal: an agent-based modeling and machine learning approach** *Quaternary Science Reviews*
R Vahdati A, Weissmann JD, Timmermann A, Ponce de León MS, Zollikofer CPE 2019, 221
- Agents.jl: agent-based modeling framework in Julia** *Journal of Open Source Software*
R Vahdati, A 2019, 4(42), 1611
- Population size affects adaptation in complex ways: simulations on empirical adaptive landscapes.** *Evolutionary Biology*
Vahdati AR, Wagner A 2017
- Effect of Population Size and Mutation Rate on the Evolution of RNA Sequences on an Adaptive Landscape Determined by RNA Folding** *International Journal of Biological Sciences*
Vahdati A R, Sprouffske K, Wagner A 2017;13(9):1138–51
- Parallel or convergent evolution in human population genomic data revealed by genotype networks** *BMC Evolutionary Biology*
Vahdati A R, Wagner A 2016
- VCF2Networks: applying Genotype Networks to Single Nucleotide Variants data** *Bioinformatics*
Dall’Olio G M, Vahdati A R, Bertranpetit J, Wagner A, Laayouni H 2014
- Genomic imprinting and genetic effects on muscle traits in mice** *BMC Genomics*
Kaerst S*, Vahdati A R*, Brockmann G, Hager R 2012
* Equal contribution

Teaching Experience

- BIO397 - Applied Machine Learning - block course** *University of Zurich*
Lecturer 2019-2022
- I have designed and delivered lectures and exercises on Julia programming language, linear algebra, multivariate calculus, machine learning algorithms, neural networks, and evaluation and improvement of machine learning models.
 - I have prepared students with no previous knowledge of machine learning to be able to design and perform complete machine learning workflows.
- An introduction to agent-based modeling** *University of Pusan, Busan, Switzerland*
Lecturer 2022
- I gave a lecture and hands-on tutorial on using Julia language for building agent-based models.
- Reproducible research - workshop** *EAWAG – Kastanienbaum, Switzerland*
Lecturer 2018
- I gave a lecture and hands-on tutorial on using Git version control software and collaborating using Github.